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(2)

Principle of Inclusion and Exclusion

As we know that the cardinality of the set P is the number of unique elements in P . It is denoted by $|P|$.

First Principle: If P and Q are disjoint sets, then

$$|P \cup Q| = |P| + |Q|.$$

(P) (Q)

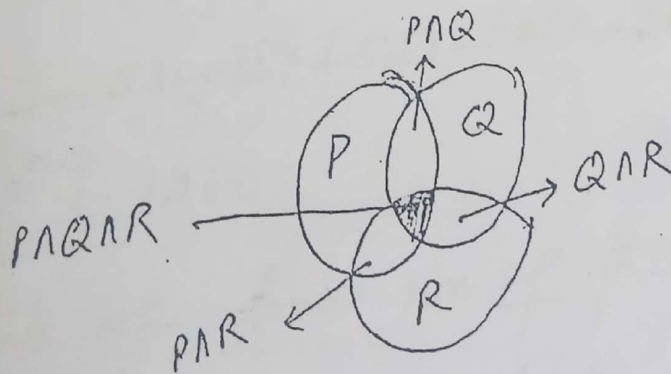
Theorem 1: Let P and Q be any two non-disjoint sets, then

$$|P \cup Q| = |P| + |Q| - |P \cap Q|.$$

(P) (Q)

Theorem 2: Let P , Q and R are three finite sets, then

$$|P \cup Q \cup R| = |P| + |Q| + |R| - |P \cap Q| - |P \cap R| - |Q \cap R| + |P \cap Q \cap R|$$



(25)

Problem 1: Out of 1200 students in a college

582 took Economics, 627 took English, 543 took Mathematics

217 took both Economics & English

307 " " Economics & Mathematics

250 " " English & Mathematics

222 took all three courses.

How many took none of the three?

Sol.: Suppose $|A| = \text{Economics} = 582$

$|B| = \text{English} = 627$

$|C| = \text{Mathematics} = 543$

$|A \cap B| = 217, |A \cap C| = 307, |B \cap C| = 250, |A \cap B \cap C| = 222$

Total number of students who took any of 3 subjects

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$= 582 + 627 + 543 - 217 - 307 - 250 + 222$$

$$= 1200$$

Students who took none of three courses

$$= \text{Total no. of students} - \text{Total students who took any of 3 subjects}$$

$$= 1200 - 1200 = 0$$

Problem 2: Among the first 500 +ve integers. Determine the integer which are not divisible by 2, nor by 3, nor by 5.

Sol:- Let A be the number of integers divisible by 2.
 B " " " " " " " 3.
 C " " " " " " " 5.

$$|A| = \left\lfloor \frac{500}{2} \right\rfloor = 250, |B| = \left\lfloor \frac{500}{3} \right\rfloor = 166, |C| = \left\lfloor \frac{500}{5} \right\rfloor = 100$$

$$|A \cap B| = \left\lfloor \frac{500}{2 \times 3} \right\rfloor = 83, |A \cap C| = \left\lfloor \frac{500}{2 \times 5} \right\rfloor = 50, |B \cap C| = \left\lfloor \frac{500}{3 \times 5} \right\rfloor = 33$$

$$|A \cap B \cap C| = 16$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

$$= 250 + 166 + 100 - 83 - 50 - 33 + 16$$

$$= 366$$

The integers not divisible by 2, 3 and 5

$$= 500 - 366 = \underline{\underline{134}}$$

Q.3: A survey of household in U.S.A reveals that 96% have at least one television set, 98% have telephone service, and 95% have television set and at least one television set. What Percentage of household in the U.S.A have neither telephone service nor a television service?

Sol.

$$|A| = \text{television set} = 96$$

$$|B| = \text{telephone service} = 98$$

$$|A \cap B| = 95$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 96 + 98 - 95 = 99\%$$

Neither telephone nor television set

$$= 100 - 99 = 1\%$$

The Pigeonhole Principle

If K is a positive integer and $K+1$ or more objects are placed into K boxes, then there is at least one box containing two or more objects.

A general principle called the Pigeonhole principle, which states that if there are more pigeons, than Pigeonholes then there must be at least one pigeonhole with at least two pigeons in it.

Following examples show, how the pigeonhole principle is used.

Example 1: Show that at least two people must have their birthday in the same month if 13 people are assembled in a room.

Solution: We assigned each person the month of the year on which he was born. Since there are 12 months in a year so, according to pigeonhole principle, there must be at least two people assigned to the same month.

Example 2: Any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthday.

Example 3: In any group of 27 english words, there must be at least two that begin with the same letter, because there are 26 letters in the english alphabet.

The generalized Pigeonhole principle

If N objects are placed into K boxes, then there is at least one box containing at least $\left\lceil \frac{N}{K} \right\rceil$ objects.

Ex: Among 100 people there are at least $\left\lceil \frac{100}{12} \right\rceil = 9$ people who share the same month.

Example: Show that if there are 30 students in a class, then at least two have last names that begin with the same letter?

Ans.

$$\left\lceil \frac{30}{26} \right\rceil = 2.$$